

Caleb Shin

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EDUCATION

University of California, Santa Cruz

Anticipated June 2028

Bachelor of Science, Electrical Engineering

Awards and Honors: Dean's List (Fall 2024)

Leadership Involvement: Formula SAE Electric, Colleges Nine College Student Life

San Diego Miramar Community College

Completed Spring 2024

Relevant Coursework: MATH 151 (Calculus II), MATH 252 (Multivariable Calculus)

San Diego Mesa Community College

Completed Summer 2024

Relevant Coursework: ENGE 151 (Computer-Aided Design)

EXPERIENCE

Vehicle Dynamics Lead, Formula Slug FSAE EV

September 2024 - Present

- Research and development of crucial physics and engineering properties of autocross vehicles
- Delegate work to subteam and advise 200-member organization on physics of vehicle design
- Created and maintained several multi-DOF suspension, steering, and tire models for gathering complex physical data to use for vehicle design
 - *Research focus in 2024:* Suspension effects on high-level vehicle handling and stability
 - *Research focus in 2025:* Analysis of tire forces at high acceleration, optimization of steering and suspension dynamics, vehicle body motion impacts on aerodynamic forces
- Present lectures on physics fundamentals and applications in automotive engineering
- Founding member of aerodynamics subteam at Formula Slug

Composites Member, Formula Slug FSAE EV

September 2024 - Present

- Research CFRP material properties and manufacture precision components
- Equip PPE and enforce all lab safety standards
- Operate and maintain vacuum systems and high-temperature curing ovens
- Manage and inventory of volatile chemicals and highly valuable materials

Residence Assistant, College Nine

August 2025 - Present

- Expedite all communication and maintain consistent availability to faculty and peers
- Organize and submit paperwork per university standards within 24 hours of reception
- Mediate interpersonal conflict and immediately secure emergency situations

PROJECTS

High DOF Automotive Steering and Suspension Model

September 2025 - Present

Major skills used: Python (numPy, matplotlib, sciPy, Jupyter), Onshape CAD, analysis of data sets

- Collection of several complicated automotive physics and geometry models, written in Python
- Fully self-driven project with almost all physics research completed independently
- Optimizing vehicle handling and response by simulating data-backed physical quantities of the vehicle, then fine-tuning design parameters to reach performance goals
- Future plans to integrate into a singular software model to simulate an entire car and its kinematics as a dynamic system

Scientific Computing and Physics Simulations

November 2025 - Present

- Independent endeavor to combine research with physical modelling and simulations to become familiar with scientific computing software through applications of fundamental and theoretical principles of electrical physics.
- Primarily interested in high-energy plasma physics, magnetohydrodynamics, and radiation embrittlement of superalloys
- Actively learning Ansys Maxwell, Fluent, and Mechanical
- Looking to gain experience with LAMMPS, RAMSES, and FLASH-X in both HPC and single-processor applications
 - Actively learning GNU, Fortran, and C++ for valuable transferrable skills in scientific computing applications